INSAIT Adobe Research

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Knowledge-Centric Agents for Workflow Generation in ComfyUI

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TL; DR

- Generating a ComfyUI workflow is **not** text-to-JSON — it is reasoning over **knowledge**: its structure, hierarchy and dynamics.
- We **invert** knowledge out of real workflows into a hierarchy, **inject** only the transitions that need learning, and **infer** by running the hierarchy backwards.
- **86.9% Pass** and **61.1% Resolve**, against 36.4% / 29.4% for the strongest

Our Methodology

Knowledge Inversion from Raw Workflows

Task Instruction → Workflow Strategy → Pseudo-Code topology → Detailed Parameters → Workflow Configuration

Train Models to Apply Knowledge

Agent Inference

Why Direct Text→JSON Fails

- **Strategic level.** Design is open-ended and experience-driven — whether to segment, how to split branches, which controls interact. That tacit expertise is not in the text corpus.
- **Structural level.** Workflows are large discrete graphs and JSON is brittle: one misconnected link or bad parameter binding and nothing runs.

→ GPT-5 zero-shot scores **0.0 on every metric** — it never produces a runnable graph.

Framework Overview

(1) Knowledge Inversion

- **Task Instruction & Inputs** — recovered user intent
- ↑ **Strategy Description** — the expert's plan
- ↑ **Strategy Reasoning** — why it is built this way
- ↑ **Skeleton Pseudo-code** — functional topology only
- ↑ **Full Pseudo-code** — modules, links, parameters
- ↑ **Raw ComfyUI Workflow** — 912 curated from 10,000+ collected

increasing abstraction ↑

(2) Knowledge Inference

- **Task Instruction & Inputs** — what the user asks for
- ↓ **High-level Strategy Planner** — SFT · + self-refinement
- ↓ **Pseudo-code Generator** — SFT · + self-refinement
- ↓ **Parameter Filler** — pretrained LLM
- ↓ **Rule-based Reconstruction** — no learning
- ↓ **Executable ComfyUI Workflow** — runs in ComfyUI

↓ generation, mirrored

Learned only where reasoning is needed. Parameter restoration and JSON reconstruction stay rule-based. An auxiliary objective — strategy reasoning extraction — makes the model spell out why each decision was taken.

152	86.9%	61.1%
Valid Node Diversity	Pass rate	Resolve rate
3.6× the best baseline (42)	ComfyAgent: 36.4%	ComfyAgent: 29.4%

Comparison on Our Test Set (30 tasks)

Method	VND↑	NodeComp↑	LinkComp↑	TaskCons↑	Pass↑	Resolve↑
GPT-5 + Zero-Shot	0	0.0	0.0	0.0	0.0	0.0
GPT-5 + Few-Shot	20	36.7	36.7	34.7	26.4	16.7
GPT-5 + CoT	26	33.3	33.3	32.9	22.8	15.8
GPT-5 + CoT-SC	21	36.7	36.7	34.8	24.2	12.5
GPT-5 + RAG	42	56.7	56.7	53.6	48.1	25.6
GPT-5 + ComfyAgent	40	56.7	56.7	55.1	36.4	29.4
Ours	**152**	**96.3**	**98.3**	**86.9**	**86.9**	**61.1**

VND distinct valid node types · **NodeComp / LinkComp** share of nodes correctly instantiated / correctly connected · **TaskCons** GPT-judged task fulfilment · **Pass** executes in ComfyUI · **Resolve** executes *and* fulfils it. All in %, except VND. Scored by running every workflow by hand. **All methods, including ours, use GPT-5 as the underlying LLM.**

Generalization to External Benchmarks

Benchmark	Best baseline (GPT-5 + RAG)	Ours
ComfyBench (20 tasks)	45 VND · 40.0 Pass · 25.0 Resolve	138 VND · 50.0 Pass · 25.0 Resolve
FlowBench (30 tasks)	42 VND · 46.7 Pass · 26.7 Resolve	145 VND · 66.7 Pass · 36.7 Resolve

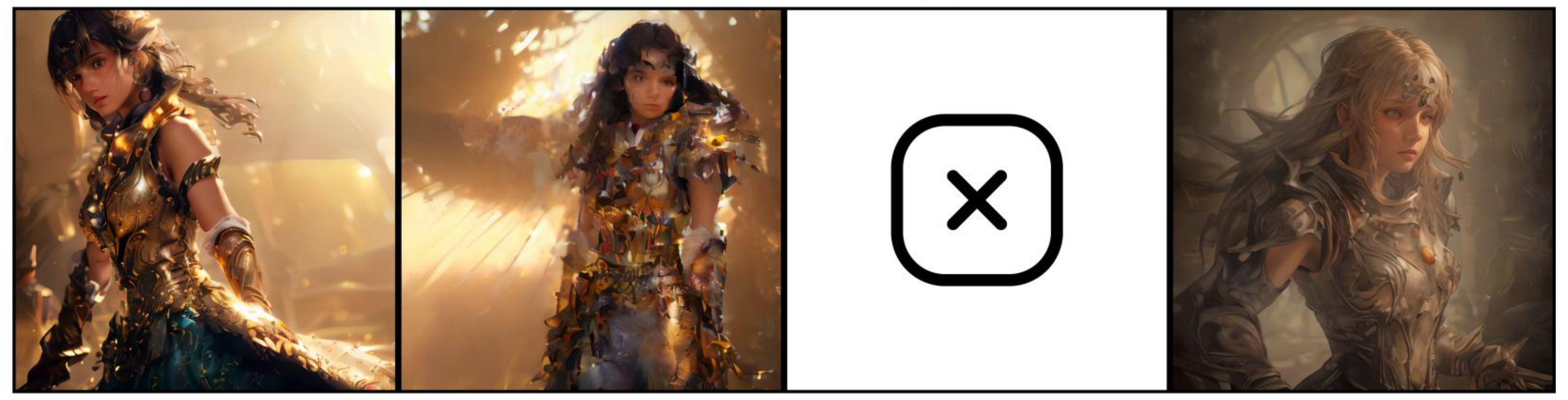
Hierarchy beats scale. Against an end-to-end Qwen3-32B (instruction→pseudo-code): Pass **86.9%** vs 56.7%, Resolve **61.1%** vs 33.3% — with a 14B backbone.

Qualitative Comparison — three tasks, same instruction given to every method

Task Instruction: Create an image from my text prompt, apply my chosen style/character presets "......" to control the look, and then upscale it with detail enhancement to a crisp, high-resolution final result suitable for large displays or print.

Task Input | GPT-5 + Few Shot | GPT-5 + RAG | ComfyAgent-GPT-5 | Ours

a hyper-detailed portrait of a person, soft natural lighting, sharp focus on eyes, realistic skin texture, cinematic depth of field...



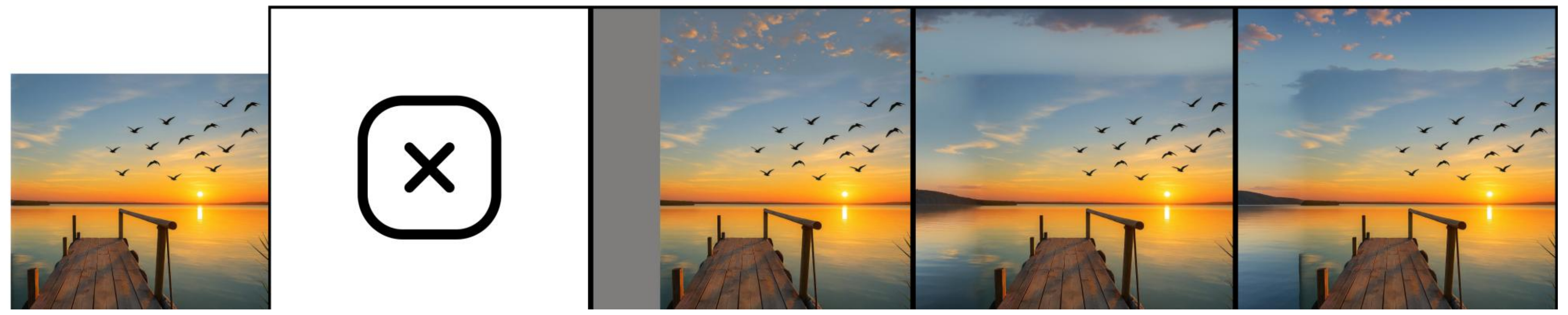
Task Instruction: Create an image from a text description using my custom style/character preset, then automatically upscale it to a much higher resolution with prompt-guided detail enhancement and save the final high-quality result.

Task Input | GPT-5 + Few Shot | GPT-5 + RAG | ComfyAgent-GPT-5 | Ours

masterpiece, best quality, ultra-detailed, steampunk airship floating above clouds, intricate gears and brass details, smoke trails, dynamic perspective, warm golden sunlight

Task Instruction: Expand my image outwards using new, coherent content that matches the style, color, and lighting of the original scene. Guide the expanded area's content according to my text prompts, maintaining the integrity and seamlessly integrating the new area.

Task Input | GPT-5 + Few Shot | GPT-5 + RAG | ComfyAgent-GPT-5 | Ours



Baselines fail to execute (✕) or silently drop part of the instruction — e.g. rendering the airship but stopping at 512×512 instead of upscaling. **Limitation:** rare node types get substituted with common ones, breaking functional dependencies.